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COURSE: COMPUTER NETWORKS

LAB: 01

Record the following TCP/IP configuration to that of others on the LAN.

IP address 172.21.20.50

Subnet Mask.....255.255.254.0

Default Gateway.....172.21.20.1

If this computer is on LAN, compare the information

A: yes

B: 255.255.254 network part is same

C: default gateway is same

D: Yes

E: 172.21.20.74

172.21.20.112

172.21.20.50

172.21.20.64

172.21.20.61

Step 3: Record the following TCP/IP information for this computer

- a. IP address: 172.21.20.50
- b. Subnet mask: 255.255.254
- c. Default Gateway: 172.21.20

Step 4: Compare this computer's TCP/IP configuration to that of others on the LAN

If this computer is on LAN, compare the information of several machines (Hosts).

- a. Are there any similarities? : yes
- b. What is similar about the IP address? : Network part is same
- c. What is similar about the default gateway? : Default Gateway is same
- d. Is there anything similar in subnet mask? :yes
- e. Record a couple of the IP addresses (of your nearby hosts)

172.21.20.74

172.21.20.112

172.21.20.50

Step 5: Check Additional TCP/IP configuration information

- a. You should see the following information:
 - a. The host name : systemlab1-10
 - b. The physical address of this machine: BC-E9-2F-A0-D5-17
 - c. IP address: 172.21.20.50
 - d. Subnet mask: 255.255.254.0
 - e. Default gateway: 172.21.20.1
 - f. DNS servers: 172.29.97.225

- b. In the LAN, compare your results with a few nearby computers. What similarities do you see in the physical (MAC) address?

Everything is same

- a. Write down the computer's host name: : systemlab1-10
- b. Write down the host name of couple of other computers

Systemlab1-11

Systemlab1-12

Systemlab1-10

Systemlab-1

Using PING TRACERT from a workstation

Ask the IP address of the nearby computers and ping. Note the result

```
C:\Users\Riphah>ping 172.21.20.74

Pinging 172.21.20.74 with 32 bytes of data:
Reply from 172.21.20.74: bytes=32 time=2ms TTL=128
Reply from 172.21.20.74: bytes=32 time=2ms TTL=128
Reply from 172.21.20.74: bytes=32 time=2ms TTL=128
Reply from 172.21.20.74: bytes=32 time=2ms TTL=128

Ping statistics for 172.21.20.74:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Users\Riphah>
```

The results are successful.

- c. Ping the IP address of default gateway and DNS server. Were the results successful?

```
C:\Users\Riphah>ping 172.21.20.1

Pinging 172.21.20.1 with 32 bytes of data:
Reply from 172.21.20.1: bytes=32 time=1ms TTL=64
Reply from 172.21.20.1: bytes=32 time=1ms TTL=64
Reply from 172.21.20.1: bytes=32 time=1ms TTL=64
Reply from 172.21.20.1: bytes=32 time<1ms TTL=64

Ping statistics for 172.21.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\Riphah>
```

- d. Ping the **Loop back ip address**. Type the following address.

```
C:\Users\Riphah>Ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Riphah>
```

- e. If loop back testing is successful then the TCP/IP is properly installed and functioning on this computer.

Yes it is functioning properly.

- f. Ping the host name of the computer that you recorded in previous part.

```
C:\Users\Riphah>ping systemlab1-10

Pinging systemlab1-10 [fe80::e51:1794:a2cc:f92a%11] with 32 bytes of data:
Reply from fe80::e51:1794:a2cc:f92a%11: time<1ms
Reply from fe80::e51:1794:a2cc:f92a%11: time<1ms
Reply from fe80::e51:1794:a2cc:f92a%11: time<1ms
Reply from fe80::e51:1794:a2cc:f92a%11: time<1ms

Ping statistics for fe80::e51:1794:a2cc:f92a%11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Riphah>
```

- g. Ping the Microsoft website www.microsoft.com

```
C:\Users\Riphah>ping www.microsoft.com

Pinging e13678.dscb.akamaiedge.net [184.26.130.100] with 32 bytes of data:
Reply from 184.26.130.100: bytes=32 time=176ms TTL=51
Reply from 184.26.130.100: bytes=32 time=176ms TTL=51
Request timed out.
Reply from 184.26.130.100: bytes=32 time=176ms TTL=51

Ping statistics for 184.26.130.100:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 176ms, Maximum = 176ms, Average = 176ms

C:\Users\Riphah>
```

Trace the route to Riphah International University's Website.

```
C:\Users\Riphah>tracert www.riphah.edu.pk

Tracing route to www.riphah.edu.pk [188.114.96.7]
over a maximum of 30 hops:

  1      *            *            *            Request timed out.
  2      *            *            *            Request timed out.
  3      6 ms        2 ms        7 ms        10.162.196.41
  4      *            5 ms        5 ms        10.58.32.250
  5      24 ms       22 ms       23 ms       static.khi77.pie.net.pk [221.120.200.3]
  6      24 ms       24 ms       24 ms       188.114.96.7

Trace complete.

C:\Users\Riphah>
```

